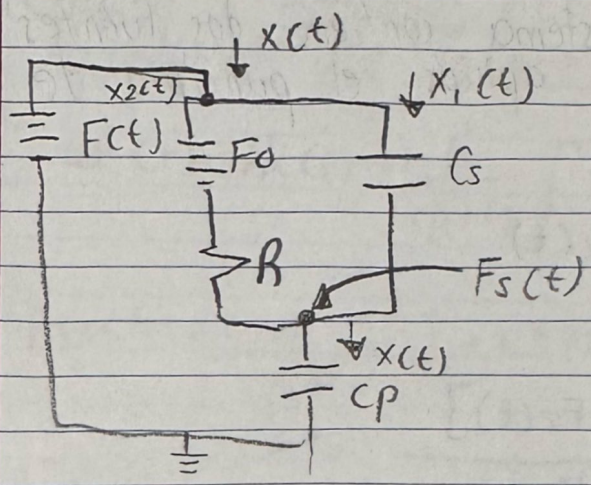
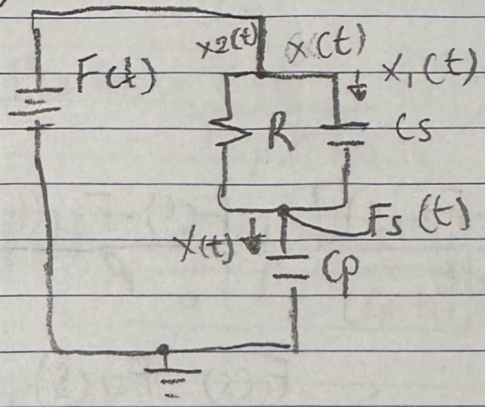
Analis matematico

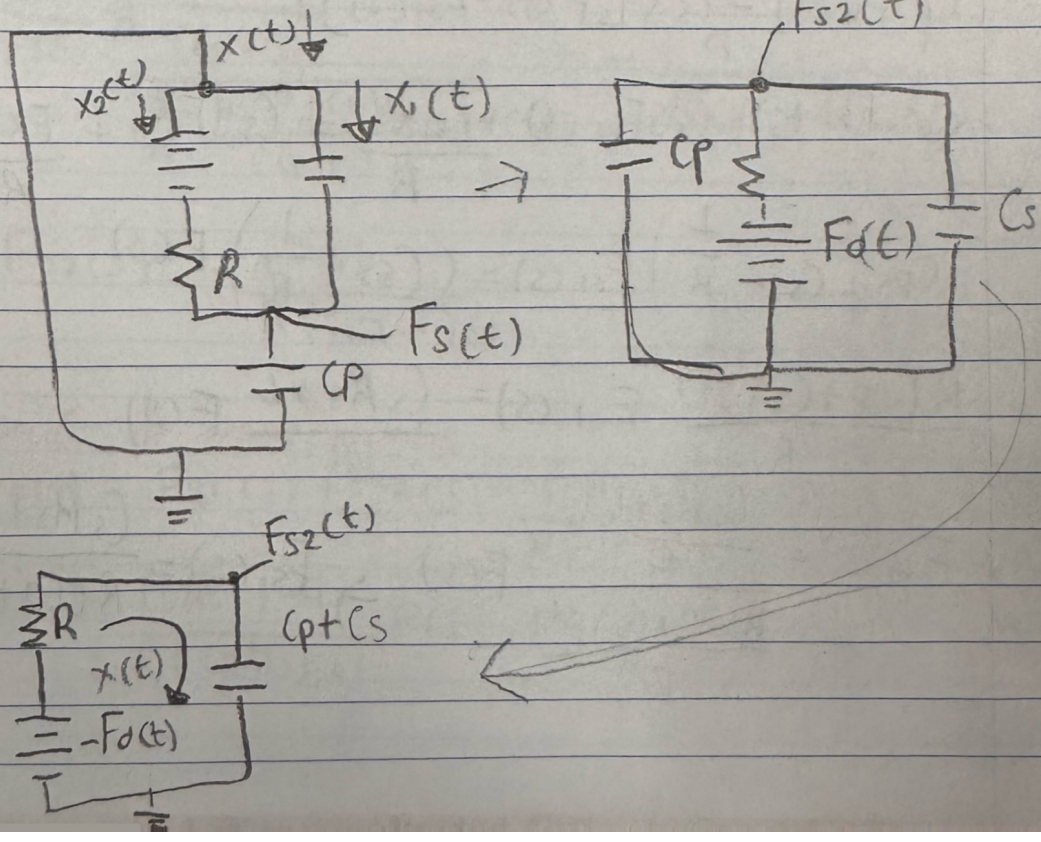
$x(t) = x_1(t) + x_2(t)$



1



2



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Debido a que el sistema contiene dos fuentes de voltaje se debe aplicar el principio de superposición

Circuito 1. Apagado

$$x(t) = x_1(t) + x_2(t)$$

$$x(t) = C_p \frac{dF_{s1}}{dt}$$

$$x_1(t) = C_s \frac{d[F(t) - F_s(t)]}{dt}$$

$$x_2(t) = \frac{F(t) - F_{s1}(t)}{R}$$

$$C_p \frac{dF_{s1}(t)}{dt} = C_s \frac{d[F(t) - F_{s1}(t)]}{dt} + \frac{F(t) - F_{s1}(t)}{R}$$

$$C_p s F_{s1}(s) = C_s s [F(s) - F_{s1}(s)] + \frac{F(s) - F_{s1}(s)}{R}$$

$$C_p s F_{s1}(s) + C_s s F_{s1}(s) + F_{s1}(s) = C_s s F(s) + F(s)$$

$$\left(C_p s + C_s s + \frac{1}{R}\right) F_{s1}(s) = \left(C_s s + \frac{1}{R}\right) F(s)$$

$$\frac{R(C_p + C_s)s + 1}{R} F_{s1}(s) = \frac{C_s R s + 1}{R} F(s)$$

$$F_{s1}(s) = \frac{\frac{C_s R s + 1}{1}}{\frac{R(C_p + C_s)s + 1}{R}} F(s) \rightarrow F_{s1}(s) = \frac{C_s R s + 1}{R(C_p + C_s)s + 1} F(s)$$

Circuito 2: Apagando F(t)
Recordando que $F_0(t) = \alpha F(t)$

$$-aF(t) = R_x(t) + \frac{1}{(C_p+C_s)} \int x(t)\,dt$$

$$F_{s2}(t) = \frac{1}{(C_p+C_s)} \int x(t)\,dt$$

$$-aF(s) = R_x(s) + \frac{x(s)}{((C_p+C_s)s}$$

$$F_{s2}(s) = \frac{x(s)}{((C_p+C_s)s}$$

$$F(s) = -\frac{1}{a}\left(R + \frac{1}{((C_p+C_s)s}\right) x(s) = -\frac{1}{a}\frac{R((C_p+C_s)s+1}{((C_p+C_s)s} x(s)$$

$$\frac{F_{s2}(s)}{F(t)} = \frac{\frac{x(s)}{((C_p+C_s)s}}{-\frac{1}{a}\frac{R((C_p+C_s)s+1}{((C_p+C_s)s} x(s)} = -\frac{a}{R((C_p+C_s)s+1}$$

$$F_{s2}(s) = -\frac{aF(s)}{R((C_p+C_s)s+1}$$

Función de transferencia

$$F_s(s) = F_{s1}(s) + F_{s2}(s) = \frac{C_s R s+1}{R((C_p+C_s)s+1} F(s) - \frac{aF(s)}{R((C_p+C_s)s+1}$$

$$F_s(s) = \frac{C_s R s+1-a}{R((C_p+C_s)s+1} F(s)$$

$$\frac{F_s(s)}{F(s)} = \frac{C_s R s+1-a}{R((C_p+C_s)s+1}$$

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Error en estado estacionario

$$e(s) = \lim_{s \to 0} sF(s)\left(1 - \frac{F_s(s)}{F(s)}\right) = \lim_{s \to 0} s\frac{1}{s}\left(1 - \frac{RC_s s + 1 - a}{(RC_p + RC_s)s + 1}\right) =$$

Estabilidad en lazo abierto

$$(RC_p + RC_s)s + 1 = 0$$

$$\lambda = -\frac{1}{RC_p + RC_s}$$

El sistema es estable